ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 23	$\frac{4325+25p}{77+p}=49$ gives $(25-49)p=-552$; $p=23$
2	**(c)** 9	$238.4=230+\frac{50-(20+x)}{25}\times 10$ gives $x=9$; $x=9$; then $y=100-82-x=9$
3	**(b)** 269.8	$\sum f_i=100$ and $\sum f_ix_i=26980$; Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{26980}{100}=269.8$
4	**(c)** 108	Median class 100–120: $l=100$, $cf=44$, $f=15$, $h=20$; Median $=l+\frac{\frac{N}{2}-cf}{f}\times h=100+\frac{50-44}{15}\times 20=108$
5	**(a)** 63	$l=60$, $f_1=24$, $f_0=21$, $f_2=17$, $h=10$; Mode $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=60+\frac{3}{10}\times 10=63$
6	**(a)** 25	$20+\frac{25-18}{14}\times 10=25$
7	**(a)** 20	$\frac{14660+190p}{80+p}=184.6$ gives $(190-184.6)p=108$; $p=20$
8	**(b)** 21	$25=20+\frac{50-(20+x)}{18}\times 10$ gives $x=21$; $x=21$; then $y=100-75-x=4$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{10+20}{2}=15$; R is the definition used.
10	**(c)** A is true but R is false.	A is the correct rule; the class with the highest frequency is the *modal* class, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\sum f_i=83+p$ and $\sum f_ix_i=23150+250p$; $\frac{23150+250p}{83+p}=274$ gives $(250-274)p=-408$ (1); $p=17$ (1).	1+1
12	Highest frequency is 23, so the modal class is 20–30; $l=20$, $f_1=23$, $f_0=4$, $f_2=4$, $h=10$ (1); Mode $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=20+\frac{19}{38}\times 10=25$ (1).	1+1
13	Cumulative frequencies: 23, 27, 52, 58, 61, 80, 100; $N=100$, so $\frac{N}{2}=50$; Median class 240–260: $l=240$, $cf=27$, $f=25$, $h=20$ (1); Median $=l+\frac{\frac{N}{2}-cf}{f}\times h=240+\frac{50-27}{25}\times 20=258.4$ (1).	1+1
14	$\sum f_i=98+p$ and $\sum f_ix_i=7730+75p$; $\frac{7730+75p}{98+p}=78.8$ gives $(75-78.8)p=-7.6$ (1); $p=2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\sum f_i=100$ and $\sum f_ix_i=7530$; Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{7530}{100}=75.3$ (1); $l=80$, $f_1=23$, $f_0=20$, $f_2=2$, $h=10$ (1); Mode: $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=80+\frac{3}{24}\times 10=81.25$ (1).	1+1+1
16	Take $a=75$, $h=10$; $u_i=\frac{x_i-a}{h}$: -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i=100$ and $\sum f_iu_i=-70$ (1); Mean $=a+h\frac{\sum f_iu_i}{\sum f_i}=75+10\times\frac{-70}{100}=68$ (1).	1+1+1
17	Median class is 230–240 with $f=20$, and the cumulative frequency before it is $cf=24+x$ (1); $239.5=230+\frac{50-(24+x)}{20}\times 10$ gives $x=7$ (1); $x=7$; then $y=100-80-x=13$ (1).	1+1+1
	OR: $\sum f_i=100$ and $\sum f_ix_i=10360$; Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{10360}{100}=103.6$ (1); $l=120$, $f_1=24$, $f_0=15$, $f_2=17$, $h=20$ (1); Mode: $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=120+\frac{9}{16}\times 20=131.25$ (1).	1+1+1
18	Take $a=270$, $h=20$; $u_i=\frac{x_i-a}{h}$: -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i=100$ and $\sum f_iu_i=-28$ (1); Mean $=a+h\frac{\sum f_iu_i}{\sum f_i}=270+20\times\frac{-28}{100}=264.4$ (1).	1+1+1
	OR: Median class is 50–60 with $f=24$, and the cumulative frequency before it is $cf=17+x$ (1); $55=50+\frac{50-(17+x)}{24}\times 10$ gives $x=21$ (1); $x=21$; then $y=100-56-x=23$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	(i) $\frac{30+40}{2} = 35$ (1). (ii) 40-50, the class with the highest frequency 14 (1). (iii) Class marks 15, 25, 35, 45, 55; $\sum fx = 30 + 100 + 350 + 630 + 550 = 1660$; mean $= \frac{1660}{40} = 41.5$ min (1). (iv) Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} h = 40 + \frac{14 - 10}{28 - 10 - 10} \times 10 = 45$ min (2).	1+1+1+2
20	Median class is 20–30 with $f = 18$, and the cumulative frequency before it is $cf = 20 + x$ (2); $25 = 20 + \frac{50-(20+x)}{18} \times 10$ gives $x = 21$ (2); $x = 21$; then $y = 100 - 75 - x = 4$ (1).	2+2+1
	OR: $\sum f_i = 100$ and $\sum f_i x_i = 5530$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{5530}{100} = 55.3$ (2); $l = 60$, $f_1 = 23$, $f_0 = 16$, $f_2 = 10$, $h = 10$ (2); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 60 + \frac{7}{20} \times 10 = 63.5$ (1).	2+2+1

Mistakes to avoid:

- Taking cf of the median class as its own cumulative frequency; cf in the formula is that of the class *before* the median class.
- Choosing the median class by comparing with n instead of $\frac{n}{2}$.
- Confusing f_0 and f_2 (preceding and succeeding class frequencies) in the mode formula.
- Using the class limits instead of the class marks x_i in $\sum f_i x_i$.
- In the step-deviation method, forgetting to multiply $\frac{\sum f_i u_i}{\sum f_i}$ by h.
- Using the modal class (highest frequency) as the median class, or vice versa.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME — SET 2

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 242	$l = 240$, $f_1 = 24$, $f_0 = 22$, $f_2 = 6$, $h = 20$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 240 + \frac{2}{20} \times 20 = 242$
2	**(b)** $20 + \frac{10}{18} \times 10$	$\frac{f_1 - f_0}{2f_1 - f_0 - f_2} = \frac{10}{32 - 14}$
3	**(d)** 25	$\frac{7890 + 110p}{75 + p} = 106.4$ gives $(110 - 106.4)p = 90$; $p = 25$
4	**(a)** 22	$70 = 60 + \frac{50 - (20 + x)}{16} \times 20$ gives $x = 22$; $x = 22$; then $y = 100 - 69 - x = 9$
5	**(a)** 74.4	$\sum f_i = 100$ and $\sum f_i x_i = 7440$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7440}{100} = 74.4$
6	**(a)** 65	Median class 60–70: $l = 60$, $cf = 49$, $f = 2$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 60 + \frac{50 - 49}{2} \times 10 = 65$
7	**(c)** 234	$l = 230$, $f_1 = 24$, $f_0 = 18$, $f_2 = 15$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 230 + \frac{6}{15} \times 10 = 234$
8	**(d)** 43	$45 + 10 \times \frac{-12}{60}$
9	**(d)** A is false but R is true.	The modal class has the *highest* frequency, so A is false; R is true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$\frac{150}{5} = 30$, so A is true; R is also a true property but it is not what gives the value of the mean.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Highest frequency is 22, so the modal class is 220–240; $l = 220$, $f_1 = 22$, $f_0 = 13$, $f_2 = 7$, $h = 20$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 220 + \frac{9}{24} \times 20 = 227.5$ (1).	1+1
12	Cumulative frequencies: 21, 34, 59, 64, 70, 94, 98, 100; $N = 100$, so $\frac{N}{2} = 50$; Median class 60–70: $l = 60$, $cf = 34$, $f = 25$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 60 + \frac{50 - 34}{25} \times 10 = 66.4$ (1).	1+1
13	$\sum f_i = 77 + p$ and $\sum f_i x_i = 4035 + 65p$; $\frac{4035 + 65p}{77 + p} = 55.3$ gives $(65 - 55.3)p = 223.1$ (1); $p = 23$ (1).	1+1
14	Highest frequency is 25, so the modal class is 100–120; $l = 100$, $f_1 = 25$, $f_0 = 13$, $f_2 = 5$, $h = 20$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 100 + \frac{12}{32} \times 20 = 107.5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\sum f_i = 100$ and $\sum f_i x_i = 18680$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{18680}{100} = 186.8$ (1); $l = 180$, $f_1 = 22$, $f_0 = 21$, $f_2 = 21$, $h = 20$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 180 + \frac{1}{2} \times 20 = 190$ (1).	1+1+1
16	Take $a = 245$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -4, -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i = 100$ and $\sum f_i u_i = -33$ (1); Mean $= a + h \frac{\sum f_i u_i}{\sum f_i} = 245 + 10 \times \frac{-33}{100} = 241.7$ (1).	1+1+1
17	Median class is 80–100 with $f = 16$, and the cumulative frequency before it is $cf = 31 + x$ (1); $93.75 = 80 + \frac{50 - (31 + x)}{16} \times 20$ gives $x = 8$ (1); $x = 8$; then $y = 100 - 76 - x = 16$ (1).	1+1+1
	OR: $\sum f_i = 100$ and $\sum f_i x_i = 6380$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{6380}{100} = 63.8$ (1); $l = 80$, $f_1 = 23$, $f_0 = 16$, $f_2 = 10$, $h = 10$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 80 + \frac{7}{20} \times 10 = 83.5$ (1).	1+1+1
18	Take $a = 270$, $h = 20$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i = 100$ and $\sum f_i u_i = 50$ (1); Mean $= a + h \frac{\sum f_i u_i}{\sum f_i} = 270 + 20 \times \frac{50}{100} = 280$ (1).	1+1+1
	OR: Median class is 250–260 with $f = 24$, and the cumulative frequency before it is $cf = 41 + x$ (1); $252.5 = 250 + \frac{50 - (41 + x)}{24} \times 10$ gives $x = 3$ (1); $x = 3$; then $y = 100 - 80 - x = 17$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{50}{2} = 25$ (1). (ii) 5, 15, 35, 43, 50 (1). (iii) 300-400, the first class whose cumulative frequency (35) exceeds 25 (1). (iv) $l = 300$, $cf = 15$, $f = 20$, $h = 100$: median $= 300 + \frac{25 - 15}{20} \times 100 = 350$ units (2).	1+1+1+2

20	$\sum f_i = 89 + p$ and $\sum f_i x_i = 9590 + 70p$ (2); $\frac{9590+70p}{89+p} = 103.6$ gives $(70 - 103.6)p = -369.6$ (2); $p = 11$ (1).	2+2+1
	OR: Take assumed mean $a = 45$; deviations $d_i = x_i - a$: -40, -30, -20, -10, 0, 10, 20, 30 (2); $\sum f_i = 100$ and $\sum f_i d_i = 400$ (2); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 45 + \frac{400}{100} = 49$ (1).	2+2+1

Mistakes to avoid:

- Taking cf of the median class as its own cumulative frequency; cf in the formula is that of the class *before* the median class.
- Choosing the median class by comparing with n instead of $\frac{n}{2}$.
- Confusing f_0 and f_2 (preceding and succeeding class frequencies) in the mode formula.
- Using the class limits instead of the class marks x_i in $\sum f_i x_i$.
- In the step-deviation method, forgetting to multiply $\frac{\sum f_i u_i}{\sum f_i}$ by h.
- Using the modal class (highest frequency) as the median class, or vice versa.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 77	$\sum f_i = 100$ and $\sum f_i x_i = 7700$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7700}{100} = 77$
2	**(b)** 96.25	Median class 80–100: $l = 80$, $cf = 37$, $f = 16$, $h = 20$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 80 + \frac{50-37}{16} \times 20 = 96.25$
3	**(d)** 231.25	$l = 220$, $f_1 = 21$, $f_0 = 3$, $f_2 = 7$, $h = 20$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 220 + \frac{18}{32} \times 20 = 231.25$
4	**(d)** 4	$\bar{x} - a$
5	**(d)** 25	$\frac{8130+70p}{75+p} = 98.8$ gives $(70 - 98.8)p = -720$; $p = 25$
6	**(d)** 14	$290 = 280 + \frac{50-(28+x)}{16} \times 20$ gives $x = 14$; $x = 14$; then $y = 100 - 66 - x = 20$
7	**(c)** 75.3	$\sum f_i = 100$ and $\sum f_i x_i = 7530$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7530}{100} = 75.3$
8	**(b)** 68.75	Median class 60–80: $l = 60$, $cf = 43$, $f = 16$, $h = 20$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 60 + \frac{50-43}{16} \times 20 = 68.75$
9	**(a)** Both A and R are true and R is the correct explanation of A.	The formula in R is an algebraic rewriting of $\sum f_i x_i / \sum f_i$, so both methods give the same mean.
10	**(c)** A is true but R is false.	The median class is found by $\frac{n}{2} = 25$: A true; the mean and median are generally different, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cumulative frequencies: 19, 30, 38, 63, 71, 85, 100; $N = 100$, so $\frac{N}{2} = 50$; Median class 70–80: $l = 70$, $cf = 38$, $f = 25$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 70 + \frac{50-38}{25} \times 10 = 74.8$ (1).	1+1
12	$\sum f_i = 95 + p$ and $\sum f_i x_i = 10210 + 110p$; $\frac{10210+110p}{95+p} = 107.6$ gives $(110 - 107.6)p = 12$ (1); $p = 5$ (1).	1+1
13	Highest frequency is 20, so the modal class is 180–200; $l = 180$, $f_1 = 20$, $f_0 = 9$, $f_2 = 15$, $h = 20$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 180 + \frac{11}{16} \times 20 = 193.75$ (1).	1+1
14	Cumulative frequencies: 10, 17, 25, 48, 73, 92, 100; $N = 100$, so $\frac{N}{2} = 50$; Median class 80–100: $l = 80$, $cf = 48$, $f = 25$, $h = 20$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 80 + \frac{50-48}{25} \times 20 = 81.6$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\sum f_i = 100$ and $\sum f_i x_i = 8840$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{8840}{100} = 88.4$ (1); $l = 100$, $f_1 = 25$, $f_0 = 16$, $f_2 = 9$, $h = 20$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 100 + \frac{9}{25} \times 20 = 107.2$ (1).	1+1+1
16	Take $a = 70$, $h = 20$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i = 100$ and $\sum f_i u_i = -71$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 70 + 20 \times \frac{-71}{100} = 55.8$ (1).	1+1+1
17	Median class is 120–140 with $f = 16$, and the cumulative frequency before it is $cf = 23 + x$ (1); $127.5 = 120 + \frac{50-(23+x)}{16} \times 20$ gives $x = 21$ (1); $x = 21$; then $y = 100 - 62 - x = 17$ (1).	1+1+1
	OR: $\sum f_i = 100$ and $\sum f_i x_i = 7620$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7620}{100} = 76.2$ (1); $l = 120$, $f_1 = 24$, $f_0 = 5$, $f_2 = 5$, $h = 20$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 120 + \frac{19}{38} \times 20 = 130$ (1).	1+1+1
18	Take $a = 270$, $h = 20$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i = 100$ and $\sum f_i u_i = 20$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 270 + 20 \times \frac{20}{100} = 274$ (1).	1+1+1
	OR: Median class is 260–280 with $f = 12$, and the cumulative frequency before it is $cf = 37 + x$ (1); $265 = 260 + \frac{50-(37+x)}{12} \times 20$ gives $x = 10$ (1); $x = 10$; then $y = 100 - 72 - x = 18$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	(i) 30 (1). (ii) $x = 70$, $u = \frac{70-50}{20} = 1$ (1). (iii) u values are $-2, -1, 0, 1, 2$; $\sum fu = -10 - 8 + 0 + 10 + 10 = 2$ (1). (iv) $\bar{x} = 50 + 20 \times \frac{2}{40} = 51$ marks (2).	1+1+1+2
20	$\sum f_i = 98 + p$ and $\sum f_i x_i = 6250 + 65p$ (2); $\frac{6250+65p}{98+p} = 63.8$ gives $(65 - 63.8)p = 2.4$ (2); $p = 2$ (1).	2+2+1
	OR: Take assumed mean $a = 130$; deviations $d_i = x_i - a$: -80, -60, -40, -20, 0, 20, 40, 60 (2); $\sum f_i = 100$ and $\sum f_i d_i = -2360$ (2); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 130 + \frac{-2360}{100} = 106.4$ (1).	2+2+1

Mistakes to avoid:

- Taking cf of the median class as its own cumulative frequency; cf in the formula is that of the class *before* the median class.
- Choosing the median class by comparing with n instead of $\frac{n}{2}$.
- Confusing f_0 and f_2 (preceding and succeeding class frequencies) in the mode formula.
- Using the class limits instead of the class marks x_i in $\sum f_i x_i$.
- In the step-deviation method, forgetting to multiply $\frac{\sum f_i u_i}{\sum f_i}$ by h.
- Using the modal class (highest frequency) as the median class, or vice versa.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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